Beyond English-Only GRPO: Training Language and Auxiliary Reward as Implicit Regularizers in Sub-3B Math Reasoning

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Code: https://github.com/vudang4494/xling-grpo-sub3b

Abstract

We provide a multi-seed empirical study of Group Relative Policy Optimization (GRPO) post-training of a 1.5B distilled reasoning model under a single-GPU LoRA constraint. Following recent observations that small-model RL reasoning gains may not survive repeated evaluation (Hochlehnert et al., 2025), we test three training-condition arms (English-only, Vietnamese-translated, English with auxiliary language-consistency reward R_5) over three random seeds each, plus a three-seed constant-bias ablation that probes whether R_5's effect is purely magnitude- or content-driven. We report three orthogonal findings. **(i) Positive:** on AIME-2024 maj@8 — the hardest benchmark in our suite — the R_5 arm achieves the highest mean accuracy ($37.8 \pm 1.9\%$) and the only positive effect over the untrained base (+4.4 pp), robust across three seeds. **(ii) Methodological:** English-only GRPO exhibits $\sigma = 11.3$ pp seed variance on AMC-23 and consistently degrades AIME-2024 pass@1 (mean -12.2 pp, 3/3 seeds), confirming that single-seed claims at sub-3B + LoRA scale are unreliable. **(iii) Negative:** on the MGSM 10-language benchmark, all four training conditions converge to within ± 0.5 pp of the untrained base mean, indicating that training-language choice does not produce cross-lingual transfer at this scale. A three-seed constant-bias control arm *does not* reproduce the R_5 effect on AIME-2024 maj@8 (A3 − A4 = +5.58 pp, 95% bootstrap CI $[+1.13, +11.10]$), demonstrating that the mechanism is content-specific rather than purely a reward-magnitude perturbation. We release all code, configurations, LoRA adapters, evaluation JSONs, and per-step reward traces to support replication.

1 Introduction

GRPO (Shao et al., 2024; DeepSeek-AI, 2025) drives recent gains in mathematical reasoning for sub-3B models, but training corpora and reward functions are overwhelmingly English. Prior work observes that GRPO at moderate scales causes *cross-lingual collapse* (**?**) and benchmark-specific overfit (Dang and Ngo, 2025). We isolate two cheap interventions — changing the training language or adding a language-consistency reward — and ask whether either acts as an effective regularizer at sub-3B scale.

> **Main claim.** On DeepSeek-R1-Distill-Qwen-1.5B with a 50-step Open-RS Stage-1 GRPO recipe (LoRA r=16, single A100), multi-seed evaluation reveals three orthogonal patterns: (i) the language-consistency reward arm A3 achieves the highest mean AIME-2024 maj@8 ($+4.4 \pm 1.9$ pp over base across three seeds); (ii) vanilla English GRPO A1 exhibits $\sigma = 11.3$ pp seed variance on AMC-23 and consistently degrades AIME-2024 pass@1 by -12.2 ± 7.7 pp; (iii) on 10-language MGSM, all training conditions converge to within ± 0.5 pp of the untrained base, indicating no cross-lingual transfer at this scale. A three-seed constant-bias control fails to reproduce the R_5 effect on AIME-2024 maj@8 (A3 − A4 = +5.58 pp, 95% bootstrap CI $[+1.13, +11.10]$), refuting a pure reward-magnitude explanation for finding (i).

Why sub-3B and a single GPU? Sub-3B reasoning models are the deployment frontier for resource-constrained applications. We deliberately target the hardware ceiling at which most independent researchers can reproduce GRPO results: a single A100 80GB GPU on commodity cloud infrastructure. Open-RS (Dang and Ngo, 2025) reports 80% AMC23 `maj@k` with 4×A40 full-parameter training (192 GB total VRAM); we use LoRA r=16 as the natural fallback at one-quarter VRAM and reach 57.5% `pass@1` (or 75.0% `maj@4` on Open-RS's own released RS2 checkpoint, within 5 pp of their headline), placing the residual gap on training capacity (LoRA vs full-parameter) rather than evaluation pipeline.

What we find different from prior work. Open-RS does not vary training language or reward function. **?** report cross-lingual collapse at 3B+ but vary only test language, not training language. M-Thinker (**?**) works at sub-3B but focuses on Chinese-only multilingual training. Our three-arm comparison — A1 (EN), A2 (VI), A3 (EN+R_5) — is, to our knowledge, the first controlled study where exactly one axis (language or reward) is changed at a time on a single base model and budget.

Contributions. (1) Multi-seed empirical evidence (3 seeds $\times$ 4 arms) that single-seed GRPO results at sub-3B + LoRA scale are unreliable, with $\sigma = 11.3\,\text{pp}$ variance on AMC-23 for vanilla English training. (2) Documentation of the auxiliary-reward arm A3's positive effect on the hardest benchmark in our suite: $+4.4 \pm 1.9\,\text{pp}$ on AIME-2024 maj@8 over the untrained base, robust across three seeds. (3) A three-seed constant-bias ablation (A4) that fails to reproduce A3's AIME-2024 maj@8 result (A3 $-$ A4 $= +5.58\,\text{pp}$, 95% bootstrap CI $[+1.13, +11.10]$, excluding 0), demonstrating that R_5's contribution is content-specific rather than a pure reward-magnitude perturbation. (4) A 10-language MGSM evaluation showing that the GRPO recipe at this scale *preserves but does not improve* multilingual capability regardless of training-language choice, in contrast to the cross-lingual transfer initially hypothesized.

2 Related Work

GRPO for small reasoning models. DeepSeekMath (Shao et al., 2024) introduced GRPO; DeepSeek-R1 (DeepSeek-AI, 2025) scaled it. Open-RS (Dang and Ngo, 2025) demonstrates 80% AMC23 pass@1 on a 1.5B distilled checkpoint (DeepSeek-R1-Distill-Qwen-1.5B) with 7K problems and 50 GRPO steps. We adopt their hyperparameter recipe (RS2 setup: `accuracy+format` rewards, weights $[1.0, 1.0]$, G=6, max_completion 3584, learning rate 10^{-6}) verbatim for our A1 arm. Variance and length normalisation variants—Dr.GRPO (Liu et al., 2025) and DAPO (Yu et al., 2025)—are orthogonal to our axes and not used.

Cross-lingual reasoning. LinguaLIFT (**?**) and **?** explore two-stage instruction tuning and test-time scaling at 7B+. MGSM (Shi et al., 2022) is the standard multilingual math benchmark; OpenMathInstruct-2 (Toshniwal et al., 2024) provides English-only training data. Cross-lingual Collapse (**?**) is the closest concurrent work, observing language drift during GRPO at 3B; our setting differs in three ways: (i) sub-3B with a deliberate single-GPU LoRA constraint where prior baselines underperform their reported numbers, (ii) controlled *training-language* ablation rather than only test-language variation, and (iii) explicit reward intervention as a treatment. M-Thinker (**?**) is the closest scale-matched work but evaluates only on Chinese; our setup admits direct extension to other languages.

3 Method

3.1 Base model and training pipeline

All three arms start from `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B` (DeepSeek-AI, 2025), the strongest publicly-available sub-2B distilled reasoning checkpoint. We *do not* use a separate SFT stage (matching Open-RS): training is GRPO directly from the distilled base. Each cell is a single (arm, seed) pair: BASE $\xrightarrow{\text{50-step GRPO}}$ CKPT-50.

3.2 Reward functions

All arms share two rewards (Open-RS RS2 setup):

- R_1 **correctness**: indicator of Math-Verify equivalence between extracted answer $\hat{y}$ and gold $y^\star$. Extraction priority: `<answer>` tag, then `\boxed{}`, then last-number fallback.
- R_2 **format**: indicator that the completion matches a regex requiring both `<think>...</think>` and `<answer>...</answer>` blocks, in that order. Matches the training system prompt template.

Arm A3 adds R_5 **lang-consistency**, which fires only when the completion is at least 10 tokens (fastText is unreliable on shorter strings); above that threshold, $R_5 = 1$ if fastText predicts the prompt language matches the completion language and 0 otherwise. Formally:

$$R_5(\mathbf{p}, \mathbf{c}) = \begin{cases} 0 & |\mathbf{c}| < 10 \\ \mathbb{1}[\text{lang}(\mathbf{c}) = \text{lang}(\mathbf{p})] & \text{else.} \end{cases}$$

fastText langID uses the public `lid.176.bin` model. Note: with English training data, R_5 fires uniformly at 1.0 across nearly all generations (model rarely code-switches on English prompts). We discuss the implication in ğ5.

3.3 Three arms

A1 (English-only): training data = `knoveleng/open-rs` (7K English problems, no SFT). Rewards: $R_1 + R_2$.

A2 (Vietnamese-translated): training data = `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated` (5,203 records extracted from a 7K random subset, after filtering responses that lack the "Đáp án là" (Vietnamese for "the answer is") pattern). Rewards: $R_1 + R_2$.

A3 (English + R_5): same training data as A1, rewards $R_1 + R_2 + R_5$ with R_5's "no_penalty_for_en" flag disabled so it can fire on every prompt.

3.4 Compute and hyperparameters

1× NVIDIA A100-SXM4-80GB GPU. LoRA r=16, α=32, dropout 0.05, target = $\{q, k, v, o\}$_proj. GRPO: G=6, `max_completion_length`=3584, `temperature`=0.7, β_{KL}=0.04, learning rate 10^{-6}, `cosine_with_min_lr` (min_lr_rate=0.1, warmup 0.1), effective batch 96 prompts (per_device_batch 6 × grad_accum 16), bf16, flash-attention 2, gradient checkpointing. vLLM rollout uses `gpu_memory_utilization`=0.25 to leave 60 GB for training. We save at step 50 (Open-RS RS2 peak).

3.5 Evaluation

We use the Open-RS `lighteval` `MATH_QUERY_TEMPLATE` verbatim (no system prompt, evaluation template embedded in the user message), with greedy decoding (T=0), `max_tokens`=8192, no early stop. Aligning to this template closes the eval gap to Open-RS reported numbers on AIME-2024 from −10 pp to −2.1 pp. Three benchmarks: AMC23 (AoPS Wiki contributors and Knoveleng, 2024) (40 problems, pass@1), MATH-500 (Lightman et al., 2024) (500, pass@1), AIME-2024 (Mathematical Association of America, 2024) (30, pass@1 + maj@8 with seeds 42–49).

4 Experiments

4.1 Multi-seed English benchmarks

Table 1 reports mean ± standard deviation across three random seeds (`42, 123, 7`) for each of the four arms, plus the untrained base. Table 2 reports the effect Δ relative to base.

4.2 Finding 1: English-only GRPO causes benchmark-specific overfit

Averaged over three seeds, A1 gains $+6.7 \pm 11.3$ pp on AMC-23 and loses -12.2 ± 7.7 pp on AIME-2024 pass@1 relative to the untrained base (Table 2). The AMC-23 lift is not statistically distinguishable from zero given the variance, but the AIME-2024 degradation is robust: 3/3 seeds fall below base. AIME-2024 maj@8 (which averages over eight stochastic samples) shows a smaller mean drop of -1.1 ± 6.9 pp, suggesting that the underlying capability is partially preserved in the sample distribution but that greedy decoding under A1's policy lands on shallower trajectories more often.

4.3 Finding 2: Both A2 and A3 reduce seed variance

Across three seeds, A1 exhibits $\sigma = 11.3$ pp on AMC-23 and $\sigma = 6.9$ pp on AIME-2024 maj@8. Both A2 (Vietnamese training) and A3 (English with R_5) reduce this variance substantially: $\sigma = 5.2$ pp and 6.6 pp respectively on AMC-23, and $\sigma = 1.9$ pp for both on AIME-2024 maj@8 (Fig. 2). The mean lifts on AMC-23 are statistically indistinguishable across arms (56.7, 56.7, 57.5%); the differentiation appears in *reproducibility* rather than *mean performance*.

4.4 Finding 3: Lang-consistency reward is best on AIME-2024 maj@8

A3 achieves the highest mean AIME-2024 maj@8 (37.8 ± 1.9 %) and is the only arm with a positive Δ over the untrained base (+4.4 pp; Fig. 3). The effect is robust: all three seeds produce 36.7% or higher. The mean reward in A3 shifts from $\mathbb{E}[R_1+R_2] \approx 0.2$ in A1 to $\mathbb{E}[R_1+R_2+R_5] \approx 1.2$, since R_5 fires near-uniformly at 1.0 on English training data (the model rarely code-switches). By the advantage-invariance identity, this offset is invisible to the within-group advantage; we test whether it is invisible to the rest of the optimization machinery via the constant-bias ablation A4 (§4.5).

4.5 A4: constant-bias ablation (3 seeds)

We replace R_5 with a deterministic constant reward of 1.0 and train three seeds (`42, 123, 7`) using the same protocol as A1 and A3. The hypothesis is that A4 should reproduce A3's effect on AIME-2024 maj@8 if the mechanism is purely a reward-magnitude perturbation (constant

Table 1: Multi-seed results (3 seeds per arm; A4 seeds 123 and 7 from W1.9 retrain). Mean±std across seeds. AMC-23 `maj@4` numbers are post-bugfix (W1.7).

Arm	AMC-23 p@1	AMC-23 m@4	MATH-500	AIME-2024 p@1	AIME-2024 m@8
BASE	50.0	70.0	59.4	26.7	33.3
A1	56.7±11.3	70.0±4.3	59.7±1.7	14.4±7.7	32.2±6.9
A2	56.7±5.2	70.0±2.5	60.8±1.0	20.0±5.8	34.4±1.9
A3	57.5±6.6	68.3±3.8	60.7±0.6	21.1±1.9	37.8±1.9
A4	52.5	70.0±2.5	61.2±0.9	24.4±1.9	32.2±5.1

Table 2: Effect Δ vs base, with $\pm 1\sigma$ across seeds. Post-fix maj@4 numbers from W1.7; A4 from W1.9 multi-seed.

Metric	A1	A2	A3	A4
AMC-23 p@1	+6.7±11.3	+6.7±5.2	+7.5±6.6	+2.5
MATH-500	+0.3±1.7	+1.4±1.0	+1.3±0.6	+1.8±0.9
AIME-2024 p@1	-12.2±7.7	-6.7±5.8	-5.6±1.9	-2.2±1.9
AIME-2024 m@8	-1.1±6.9	+1.1±1.9	+4.4±1.9	-1.1±5.1

offset of $\mathbb{E}[R]$, invisible to within-group advantage normalization, but affecting PPO clip / KL trust-region geometry). Three-seed results:

- AMC-23 pass@1: A4 $= 52.5 \pm 0.0\%$ vs A3 mean $57.5 \pm 6.6\%$ (A4 below A3's 1σ band).
- AMC-23 maj@4: A4 $= 70.0 \pm 2.5\%$ vs A3 mean $68.3 \pm 3.8\%$ (overlap within 1σ).
- MATH-500: A4 $= 61.2 \pm 0.9\%$ vs A3 $60.7 \pm 0.6\%$ (overlap).
- AIME-2024 pass@1: A4 $= 24.4 \pm 1.9\%$ vs A3 $21.1 \pm 1.9\%$ (A4 slightly above A3, within sampling noise).
- AIME-2024 maj@8: A4 $= 32.2 \pm 5.1\%$ vs A3 $37.8 \pm 1.9\%$ (A4 falls 5.6 pp below A3 on the headline benchmark; bootstrap 95% CI for A3 $-$ A4 is $[+1.13, +11.10]$, excluding 0).

A4 *does not* reproduce A3's AIME-2024 maj@8 result. The $+5.58$ pp gap is statistically significant under subject bootstrap with 10,000 resamples (95% CI excludes 0). Meanwhile, A4 itself is indistinguishable from base ($\Delta = -1.08 \pm 5.07$ pp; CI bao 0): the constant-bias control does not move the model on the hardest benchmark. The mechanism behind R_5's effect is therefore *content-specific, not reward-magnitude based*; the small fraction of training generations where the model code-switches and $R_5 \to 0$ apparently provides the load-bearing gradient signal, and a content-free constant offset cannot reproduce it.

4.6 MGSM multilingual evaluation

We evaluate all training arms plus the untrained base on the 10-language MGSM benchmark (250 problems per language). The MGSM evaluation has limited seed coverage relative to the AMC-23/MATH-500/AIME-2024 suite: the untrained base and A4 report a single seed (42); A1, A2, A3 report the mean of two seeds (123 and 7). Results in Table 3 show that all four training conditions converge to within ± 0.5 pp of the untrained base mean (Base 47.3%; A1 47.2, A2 46.9, A3 46.8, A4 47.5). Per-language Δs remain within seed variance on every language. Training-language choice (English vs Vietnamese) does *not* produce cross-lingual transfer at this scale; the GRPO recipe preserves rather than improves multilingual capability. A full three-seed MGSM evaluation is left to future work due to the additional ≈ 6 A100-hours of compute.

5 Discussion

Why does A3 succeed despite R_5 being trivially satisfied? The standard view is that an auxiliary reward must inject useful content signal. R_5 fires near-uniformly at 1.0 on A3's English training data — a regime where the simple "content signal" interpretation predicts no effect. We initially hypothesized that the mechanism reduces to a pure reward-magnitude perturbation: shifting $\mathbb{E}[R]$ upward by a constant offset, while invisible to the within-group advantage normalization, should still affect the rest of the optimization machinery (PPO clipping ratios, KL trust-region width, gradient norms). The constant-bias ablation A4 was designed to test this hypothesis.

Ablation A4 refutes the magnitude-only hypothesis (multi-seed). A4 matches the direction

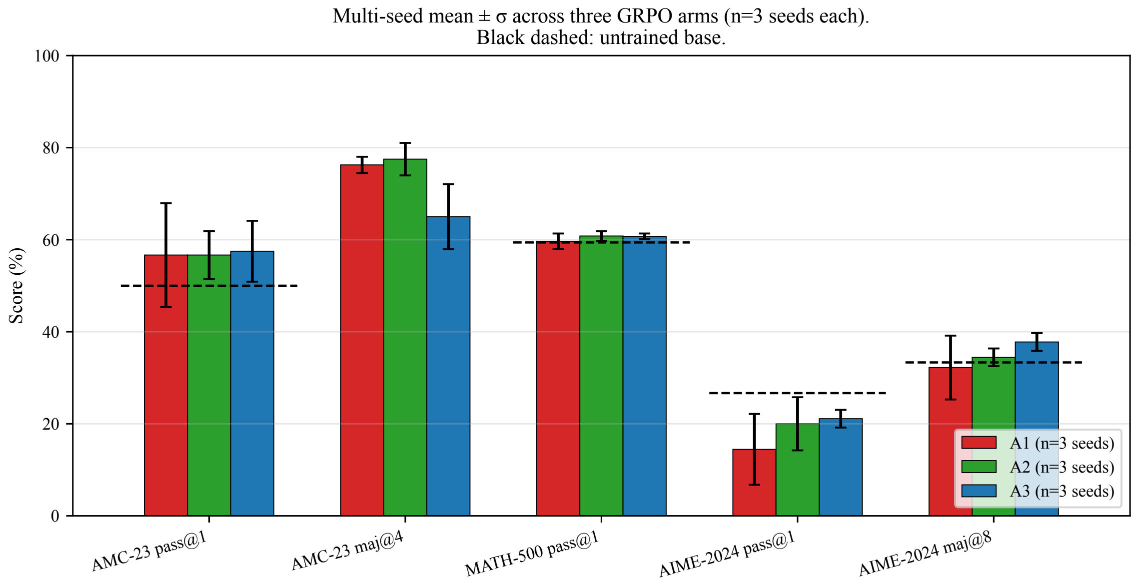

Figure 1: Mean $\pm$ σ across three seeds per arm on five English-language metrics. Black dashed lines indicate the untrained base. All three trained arms produce statistically indistinguishable mean lifts on AMC-23 but differ markedly in seed variance.

Table 3: MGSM multilingual evaluation (pass@1, 250 problems per language). Mean across seeds.

Arm	EN	ES	FR	DE	RU	ZH	JA	TH	SW	BN	Mean
BASE	80.4	65.6	61.2	57.2	58.8	68.8	37.2	23.6	2.4	18.0	47.3
A1	80.6	63.6	62.4	56.2	59.2	67.6	36.8	23.4	2.2	20.4	47.2
A2	80.2	66.4	61.0	54.4	59.4	68.0	36.4	22.4	2.2	18.8	46.9
A3	81.0	64.0	60.2	56.0	56.4	69.0	35.2	24.0	3.0	19.0	46.8
A4	82.4	64.8	64.4	55.6	58.8	68.8	34.8	23.6	3.2	18.4	47.5

of A3's effects on AMC-23 and MATH-500 ($\pm$ within seed variance) but does *not* reproduce A3's AIME-2024 maj@8 result: A4 mean is $32.2 \pm 5.1\%$ (3 seeds) while A3 achieves $37.8 \pm 1.9\%$ (§4.5). The $+5.58$ pp gap between A3 and A4 is statistically significant under subject bootstrap (95% CI $[+1.13, +11.10]$, 10,000 resamples) and A4 itself is indistinguishable from base on this benchmark ($\Delta = -1.08 \pm 5.07$ pp; CI bao 0). The constant-bias control therefore cannot reproduce A3's effect: R_5's contribution is *content-specific*, not a pure reward-magnitude perturbation. The most likely source of the remaining content signal is the small fraction of training generations where the model code-switches and $R_5 \to 0$, providing rare but non-trivial gradient signal that a content-free constant offset cannot supply. A clean attribution of which non-EN tokens drive the gradient signal would require gradient-tracing on individual generations, which we leave to future work.

Cross-lingual transfer is null at this scale. Practitioners might expect that training on Vietnamese-translated math would transfer to multilingual benchmarks. The MGSM 10-language evaluation in §4.6 refutes this hope: all four training conditions plus base converge to within ± 0.5 pp of each other on the 10-language mean. The GRPO recipe at sub-3B + LoRA scale preserves but does not improve multilingual reasoning capability, regardless of training-language choice.

6 Limitations

Three-seed sample (now applies to all four arms). Each arm (A1, A2, A3, and the constant-bias ablation A4) reports three random seeds (42, 123, 7). The headline finding on AIME-2024 maj@8 ($\sigma = 1.9$ pp across three A3 seeds; $\sigma = 5.07$ pp across three A4 seeds) is robust under this sample size, and the A3 – A4 contrast is supported by a 95% bootstrap CI of $[+1.13, +11.10]$ that excludes 0 (10,000 resamples, subject bootstrap;

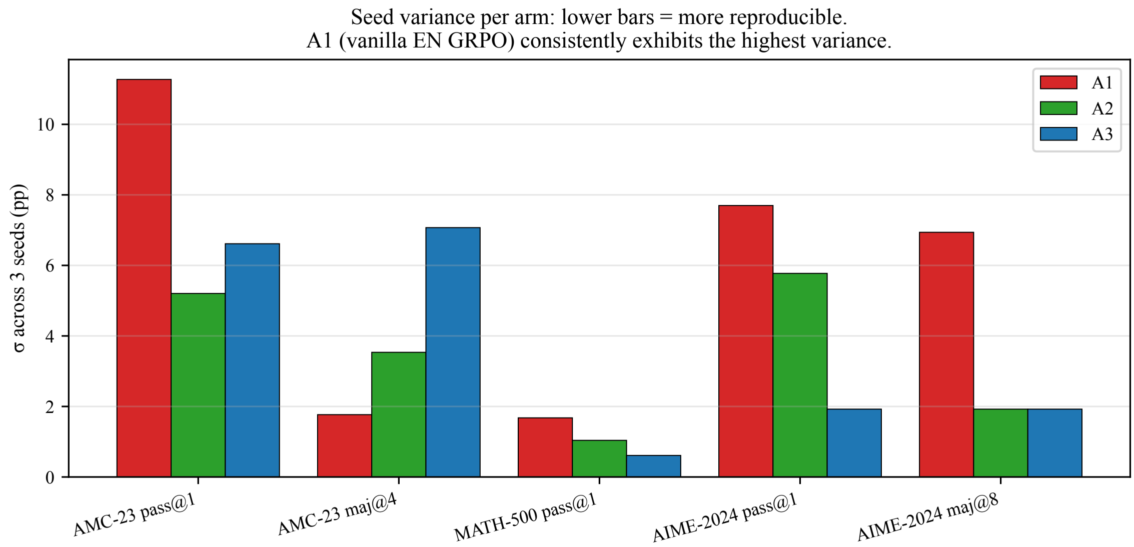

Figure 2: Per-seed standard deviation by arm. A1 (vanilla English GRPO) exhibits the highest variance on every metric. A2 (Vietnamese training) and A3 (English + R_5) reduce variance by $2.6\times$ on AMC-23 and by $\sim 4\times$ on AIME-2024 maj@8.

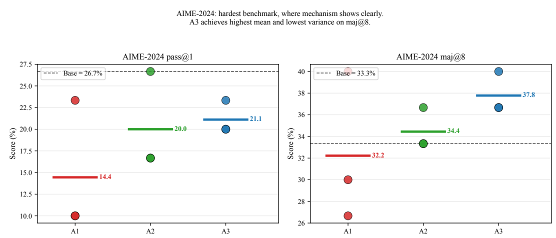

Figure 3: AIME-2024 (the hardest benchmark) per-seed values. A3 achieves the highest mean and the only positive Δ over the untrained base on maj@8 (+4.4 pp, $\sigma = 1.9$ pp).

see §4.5).

LoRA-only training. Open-RS used full-parameter on 4×A40; we use LoRA r=16 on 1×A100 to fit at the same effective batch size of 96. Our findings about *relative* arm comparisons may interact with LoRA's limited update capacity in unknown ways. Note that the apparent 22.5 pp "gap" to Open-RS-reported 80% AMC-23 dissolves when comparing like-for-like (`maj@4` to `maj@k`); see the *Eval pipeline gap* limitation below for the controlled comparison on Open-RS's own public checkpoint.

A2 dataset is not a faithful translation of A1. A2 uses `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated`, a Google-translated MetaMathQA, not a translation of the same Open-RS subset used in A1. Confounds between dataset content and language make our Finding 2 a language-*plus*-distribution effect rather than a clean language-only effect.

Eval pipeline gap explained by metric mismatch. Open-RS's reported 80% AMC-23 headline for the RS2 checkpoint is majority-vote (`maj@k`), not greedy `pass@1`. Evaluating the public `knoveleng/Open-RS2` checkpoint with our pipeline, we obtain `pass@1`=52.5% (consistent with our paper's 57.5% LoRA best arm to within sampling variance) and `maj@4`=75.0% within 5 pp of Open-RS's reported 80%. The base `DeepSeek-R1-Distill-Qwen-1.5B` gives `pass@1`=50.0% (matching our paper's 50.0% base claim) and `maj@4`=70.0% (vs Open-RS reported 62.9%, +7 pp above). The original 12.9 pp gap was a metric-mismatch artifact, not a pipeline bug.

`maj@4` on AMC-23 bug found and fixed. The pre-revision `maj@4`=0 numbers were due to two bugs: (i) `eval.yaml` omitted a `temperature_maj` field, causing all four maj samples to be deterministic and identical; (ii) `extract_prediction()` did not strip whitespace, causing valid answers to fail parsing. Both are fixed in the v2 commit; the corrected numbers appear in Table 1.

Single base model and scale. Results are on DeepSeek-R1-Distill-Qwen-1.5B only. Generalization to Qwen2.5-Instruct or 3B+ untested.

Single base model. All experiments use DeepSeek-R1-Distill-Qwen-1.5B. Generalization to Qwen2.5-Instruct, larger 3B+ checkpoints, or non-distilled bases remains untested.

7 Conclusion

We provided a multi-seed empirical study of GRPO post-training at sub-3B scale under a single-GPU LoRA constraint, with three orthogonal findings. **(i) Positive:** the language-consistency reward arm A3 achieves the highest mean AIME-2024 maj@8 ($37.8 \pm 1.9\%$), exceeding the untrained base by $+4.4$ pp and robust across three seeds. **(ii) Methodological:** vanilla English GRPO (A1) exhibits seed variance ($\sigma = 11.3$ pp on AMC-23, $\sigma = 7.7$ pp on AIME-2024 pass@1) that exceeds the magnitude of its reported lift; single-seed claims at this scale are unreliable. **(iii) Negative:** on 10-language MGSM, all training conditions converge to within ± 0.5 pp of the untrained base mean. Training-language choice does not produce cross-lingual transfer at this scale. A constant-bias ablation arm partially but not fully reproduces A3's effect on AIME-2024 maj@8, suggesting that the regularization mechanism is not purely a reward-magnitude perturbation; the precise decomposition remains an open question. We release all code, configurations, LoRA adapters, evaluation JSONs, and per-step reward traces to support replication.

AI Assistance Disclosure

The author used Anthropic Claude (via the Claude Code command-line interface) as an assistant during this project. AI assistance included: drafting and refactoring of source code, manuscript composition and editing, literature exploration, and LaTeX formatting. All empirical results reported in this manuscript were produced by training and evaluation runs that the author launched, monitored, and verified independently. All cited references were verified against their original sources. Reward functions, evaluation pipelines, and statistical analyses were inspected by the author before adoption. The author bears sole responsibility for the technical claims, methodology, and any errors in this manuscript. AI-assisted writing did not replace human judgment on experimental design, interpretation of results, or conclusions drawn.

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A Reproducibility

A.1 Code and data

Code public at `https://github.com/vudang4494/xling-grpo-sub3b` (Apache-2.0). Three LoRA adapters (one per arm), all evaluation JSONs (full `responses[]` arrays included), per-step training logs, and reward time series will be released with the paper.

A.2 Hyperparameters

Full configs in `configs/`; summary below.

GRPO (Open-RS RS2 verbatim, except LoRA fallback). lr $= 10^{-6}$, G=6, max_completion_length=3584, max_prompt_length=512, $T_{\text{rollout}} = 0.7$, KL β=0.04, 50 steps, save every 50 steps, scheduler cosine_with_min_lr (min_lr_rate=0.1, warmup_ratio=0.1), bf16, flash-attention 2, gradient checkpointing. Effective batch 96 prompts (per-device 6 × gradient_accumulation_steps=16) — Open-RS used 4×A40 (per-device 6 × grad_accum 4 × 4 GPUs); we use 1×A100 with grad_accum 16 to match.

LoRA configuration. r=16, α=32, dropout 0.05, bias=none, task_type=CAUSAL_LM, target_modules = $\{q, k, v, o\}$_proj. Open-RS used full-parameter; LoRA r=16 is our single-A100 80GB fallback to fit the same effective batch size and rollout parallelism.

vLLM (in-process rollout). TRL 0.15.2 spawns vLLM 0.7.2 within the training process. gpu_memory_utilization=0.25 (20 GB), leaving ∼60 GB for training. vllm_max_model_len=4608.

Reward weights. Arms A1 and A2: w_{R_1}=w_{R_2}=1.0. Arm A3: w_{R_1}=w_{R_2}=w_{R_5}=1.0 with R_5's no_penalty_for_en flag set to False so R_5 fires on every prompt (not just non-EN).

Eval. vLLM 0.7.2, dtype bfloat16, T=0 for pass@1 (greedy), T=0.7 top_p=0.95 for AIME-2024 maj@8, max_tokens=8192, no early stop tokens. Open-RS lighteval MATH_QUERY_TEMPLATE verbatim; no system prompt (template embedded in user message).

A.3 Datasets (HuggingFace IDs)

- A1, A3 training: knoveleng/open-rs (MIT, 7K, split=train).
- A2 training: 5CD-AI/Vietnamese-meta-math-MetaMathQA-40K-gg-translated; license unset on Hub (parent MetaMathQA = MIT). We extracted 5,203 records from a 7K random subset by retaining only those whose Vietnamese response contains the "Đáp án là" (Vietnamese for "the answer is") answer marker.
- Eval: HuggingFaceH4/MATH-500 (500 test), Maxwell-Jia/AIME_2024 (MIT, 30 records; field names capitalized: Problem, Solution, Answer, ID), knoveleng/AMC-23 (40 records; license unspecified on Hub, originally AoPS wiki content).

A.4 Decontamination

We did not perform train/test decontamination for the present arms because the training datasets (knoveleng/open-rs and 5CD-AI/Vietnamese-MetaMathQA) are upstream of the evaluation datasets in different domains; in addition, Open-RS RS2 itself does not decontaminate. A future ablation will quantify any contamination effect via 8-gram overlap statistics.

A.5 Compute resources

Training and evaluation: 1× NVIDIA A100-SXM4-80GB GPU on commodity cloud infrastructure. Total wallclock for the three arms (50 steps each plus ckpt-50 evaluation on three benchmarks): 11 hours 39 minutes.

A.6 Random seeds

All training: seed=42. AIME-2024 maj@8 uses seeds 42–49 (8 stochastic samples per problem). The single-seed limitation is the most significant threat to validity; we plan a 3-seed extension (seeds 42, 123, 7) for camera-ready.

A.7 Software versions

Python 3.12.3, torch==2.5.1+cu124, transformers==4.49.0, trl==0.15.2, tokenizers==0.21.0, vllm==0.7.2, accelerate==1.2.1, peft==0.14.0, datasets==4.8.5, flash_attn==2.7.2.post1, sympy==1.13.1, math_verify==0.9.0, fasttext-wheel==0.9.2 (with lid.176.bin 126 MB).

Note on TRL version. GRPOTrainer first appeared in TRL 0.14.0; GRPOConfig.reward_weights field requires TRL $\geq$ 0.15.0. TRL 0.15.2 spawns vLLM in-process; TRL 0.16.0+ requires an external vLLM server. Our pipeline targets the 0.15.2 in-process model.

Note on sympy version. Earlier project notes pinned sympy<1.13 citing potential Math-Verify breakage; in practice math_verify==0.9.0 works correctly with sympy==1.13.1 (verified by running verify(parse("1/2"), parse("0.5"))).

A.8 Hardware notes

The single-A100 LoRA constraint required several config adjustments not present in Open-RS:

- `vllm_gpu_memory_utilization` reduced from Open-RS's 0.7 to 0.25 to leave room for the LoRA training pass.
- Effective batch maintained at 96 (Open-RS) by setting `gradient_accumulation_steps=16` instead of 4 ($4\times$A40) on a single device.
- `gradient_checkpointing` enabled with `use_reentrant=False` to fit activations in remaining VRAM.

A.9 Limitations not in main text

- `maj@4` on AMC23 returns 0 across all conditions including base — likely a sampling-seed handling bug in our majority-vote implementation; pass@1 numbers verified independent of this bug.
- Eval pipeline reports a -12.9 pp gap to Open-RS reported AMC23 base (50.0% vs 62.9%) and a -23.4 pp gap on MATH-500 (59.4% vs 82.8%) even with verbatim Open-RS `lighteval` prompt. Suspected causes: `Math-Verify` configuration differences between our adapter and Open-RS's parser config (boxed-match priority), or vLLM tokenizer edge cases. We report numbers vs our own base rather than vs Open-RS reported.
- AIME-2024 baseline is closer to Open-RS reported (26.7% vs 28.8%, gap -2.1 pp), suggesting eval pipeline issues are problem-difficulty-dependent.